JAMA Network Open

Original Investigation | Diabetes and Endocrinology

Effectiveness of a Community-Based Structured Physical Activity Program for Adults With Type 2 Diabetes

A Randomized Clinical Trial

Aishee B. Mukherji, BA; Di Lu, MS; FeiFei Qin, MPH; Haley Hedlin, PhD; Neil M. Johannsen, PhD; Sukyung Chung, PhD; Yukari Kobayashi, MD; Francois Haddad, MD; Cynthia Lamendola, RN; Marina Basina, MD; Ruth Talamoa, BS; Jonathan Myers, PhD; Latha Palaniappan, MD, MS

Abstract

IMPORTANCE The efficacy of physical activity interventions among individuals with type 2 diabetes has been established; however, practical approaches to translate and extend these findings into community settings have not been well explored.

OBJECTIVE To test the effectiveness of providing varying frequencies of weekly structured exercise sessions to improve diabetes control.

DESIGN, SETTING, AND PARTICIPANTS The IMPACT (Initiate and Maintain Physical Activity in Communities Trial) study was a controlled randomized clinical trial (randomization occurred from October 2016 to April 2019) that included a 6-month, structured exercise intervention either once or thrice weekly vs usual care (UC; advice only). The exercise intervention was conducted at community-based fitness centers. Follow-up visits were conducted in a university research clinic. Participants included adults with type 2 diabetes (hemoglobin A_{1c} [HbA_{1c}] 6.5%-13.0%, not taking insulin, and no precluding health issues). Data analysis was performed from January to April 2022.

INTERVENTIONS A once-weekly structured exercise group, a thrice-weekly structured exercise group, or UC.

MAIN OUTCOMES AND MEASURES The primary outcome was HbA_{1c} at 6 months.

RESULTS A total of 357 participants (143 women [40.1%]) with a mean (SD) age of 57.4 (11.1) years were randomized (119 each to the UC, once-weekly exercise, and thrice-weekly exercise groups). There was no significant difference in HbA_{1c} change by study group in the intention-to-treat analysis at 6 months. Specifically, HbA_{1c} changed by −0.23% (95% CI, −0.48% to 0.01%) in the thrice-weekly exercise group and by −0.16% (95% CI, −0.41% to 0.09%) in the once-weekly exercise group. A total of 62 participants (52.1%) in the once-weekly exercise group and 56 participants (47.1%) in the thrice-weekly exercise group were at least 50% adherent to the assigned structured exercise regimen and were included in the per-protocol analysis. Per-protocol analysis showed that HbA_{1c} changed by −0.35% (95% CI, −0.60% to −0.10%; P = .005) at 3 months and by −0.38% (95% CI, −0.65% to −0.12%; P = .005) at 6 months in the thrice-weekly exercise group compared with UC. There was no significant decrease in HbA_{1c} in the once-weekly exercise group. The exercise intervention was effective in improving self-reported minutes of metabolic equivalent tasks per week for participants in the thrice-weekly exercise group (both overall and per protocol).

CONCLUSIONS AND RELEVANCE Although the intervention was not effective in the intention-to-treat analysis, participants in the thrice-weekly exercise group who attended at least 50% of the

(continued)

Key Points

Question What is the relative effectiveness of offering structured exercise sessions in improving diabetes control?

Findings In this 3-group randomized clinical trial, 357 adults with type 2 diabetes were advised to follow American Diabetes Association physical activity guidelines and randomized to usual care or once-weekly vs thrice-weekly structured exercise. There was no significant difference in hemoglobin A_{1c} in the intention-to-treat analysis, and hemoglobin A_{1c} was lowered only for participants in the thrice-weekly structured exercise program who attended at least 50% of the recommended exercise sessions.

Meaning These findings suggest that future efforts should focus on improving adherence to thrice-weekly structured exercise programs to meet exercise guidelines.

+ Visual Abstract

+ Supplemental content

Author affiliations and article information are listed at the end of this article.

Open Access. This is an open access article distributed under the terms of the CC-BY License.

Downloaded from jamanetwork.com by guest on 09/23/2026

Abstract (continued)

sessions during the 6-month exercise intervention program improved HbA_{1c} levels at 6 months. Future efforts should focus on improving adherence to thrice-weekly structured exercise programs to meet exercise guidelines.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: NCT02061579

JAMA Network Open. 2022;5(12):e2247858. doi:10.1001/jamanetworkopen.2022.47858

Introduction

The US Centers for Disease Control and Prevention have reported that an estimated 37.3 million adults (11.3% of the adult population) in the US have type 2 diabetes (T2D).[1] According to the most recent American Diabetes Association (ADA) lifestyle management guidelines, individuals with T2D should perform at least 150 minutes of moderate-to-vigorous intensity aerobic exercise per week, spread over at least 3 days, and 2 to 3 sessions of resistance training weekly.[2] Despite these ADA recommendations, approximately 38% of adults with T2D are considered inactive (less than 10 minutes of weekly physical activity), and only 24% meet the current physical activity recommendations.[3]

The value of exercise for glycemic control has been well-established in highly controlled research settings.[4-9] Previous randomized clinical trials have shown the positive effect of combined aerobic exercise and resistance training on glycemic control in T2D.[10,11] For example, Church et al[10] demonstrated that thrice-weekly structured exercise sessions including aerobic and resistance training improved hemoglobin A_{1c} (HbA_{1c}) levels compared with the nonexercise group (−0.34%; 95% CI, −0.64% to −0.03%; $P = .03$), which was not achieved by aerobic exercise or resistance exercise training alone.

Despite the known benefits of regular physical activity for patients with T2D,[10-14] physician-delivered physical activity advice alone was not effective in increasing physical activity.[15] In previous studies[10,11,16,17] of individuals with T2D, exercise adherence rates were high (80%) in highly structured research settings, but adherence to recommended physical activity (46%)[16] and follow-up attendance rates observed in general population settings were low (50%).[18,19]

In the Initiate and Maintain Physical Activity in Communities Trial (IMPACT) study, we evaluated the effectiveness of a structured exercise program either thrice weekly or once weekly compared with usual care (UC), which consisted of advice only and not taking part in 6-month exercise intervention, in improving HbA_{1c} levels. All participants were instructed to follow the ADA lifestyle management guidelines, with at least 150 minutes of moderate-to-vigorous intensity aerobic exercise per week, spread over at least 3 days, and 2 to 3 sessions of resistance training weekly that encompasses all major muscle groups. We also evaluated whether self-reported physical activity, measured by the Modifiable Activity Questionnaire, and quality of life improved within the 3 groups.

Methods

Participants and Screening

Participants for the IMPACT study were recruited from the greater San Francisco, California, Bay Area from 2015 to 2019. A complete rationale and methods have been published elsewhere.[20] Participants were screened and then randomized after completion of informed written consent and approval by their primary care physician. The Stanford University institutional review board approved this research protocol (Supplement 1). The study follows the Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines for randomized clinical trials.[21] The study was monitored annually by a data and safety monitoring board.

Downloaded from jamanetwork.com by guest on 09/23/2026

Eligibility Criteria

Individuals with T2D not taking insulin were randomized (**Figure 1**). The primary inclusion criteria were age 18 to 80 years (inclusive), diagnosis of T2D without the use of insulin, and HbA_{1c} level of 6.5% to 13.0% (to convert HbA_{1c} to proportion of total hemoglobin, multiply by 0.01). Notable exclusion criteria included body mass index (BMI; calculated as weight in kilograms divided by height in meters squared) greater than 70, HbA_{1c} greater than 13.0%, and any serious medical condition that would contraindicate long-term participation. Race and ethnicity were self-identified by the participants and were assessed in this study to facilitate generalizability assessment and to determine whether intervention effectiveness varied by these factors.

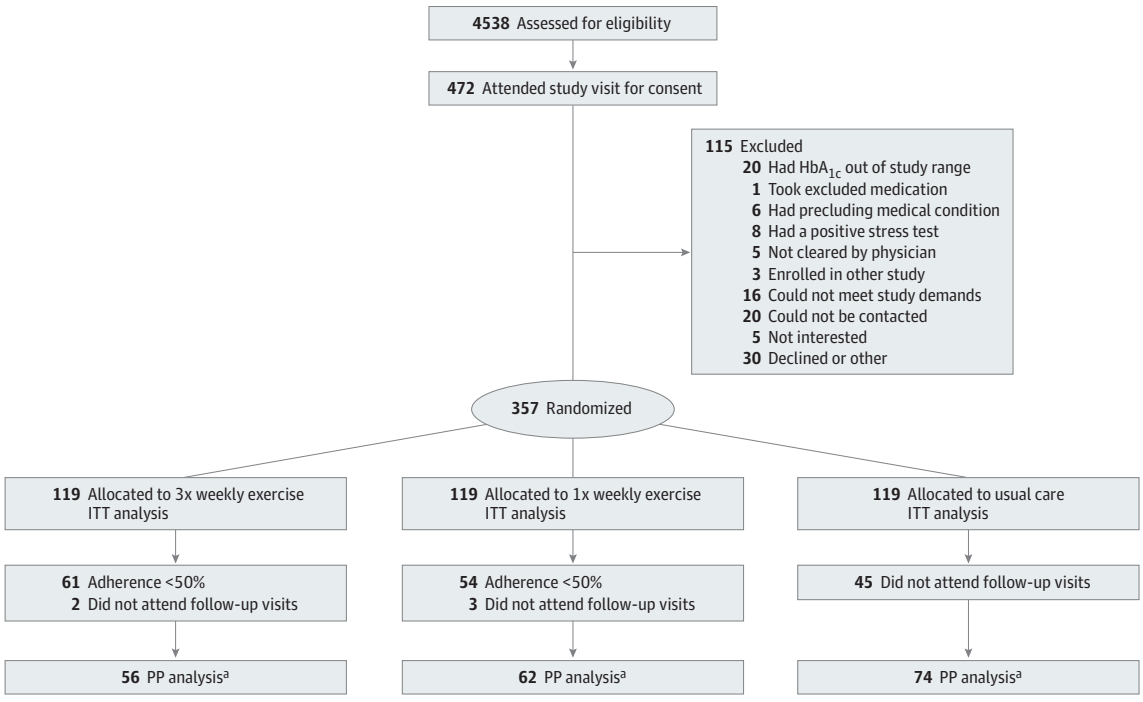
Study Design

The IMPACT study was a controlled, assessor-blinded, randomized clinical trial using a parallel-group study design with 3 study groups: (1) structured exercise 3 times per week, (2) structured exercise once per week, and (3) UC. Randomization was stratified by baseline HbA_{1c} (low, 6.5%-7.9%; high, 8.0%-9.9%) and age (young, 18.0-49.9 years; old, 50.0-80.0 years) using a block design procedure. All participants were instructed to follow ADA exercise guidelines[22] and to continue to seek UC from their physicians and were invited to attend monthly group meetings to receive information on healthy lifestyles in diabetes. The exercise groups received similar exercise protocols except for offered session frequency.

Exercise Interventions

The structured exercise sessions were conducted at more than 60 community-based fitness centers (eg, 24 Hour Fitness, YMCA) in the greater San Francisco Bay Area, covering a geographic range of more than 50 miles. Exercise training staff conducted exercise evaluations every 2 months to adjust

Figure 1. Study Enrollment Flowchart



HbA_{1c} indicates hemoglobin A_{1c}; ITT, intention-to-treat; PP, per-protocol.

[a] PP is defined as participants who attended at least 11 of 22 expected sessions in the once-weekly exercise group or at least 33 of 66 expected sessions in the thrice-weekly exercise group and attended at least 1 follow-up visit at 3 or 6 months.

Downloaded from jamanetwork.com by guest on 09/23/2026

exercise recommendations. Participants in the once-weekly exercise group were instructed to attend 1 structured combination exercise session per week (26 sessions total), whereas participants in the thrice-weekly exercise group were instructed to attend 2 structured combination exercise sessions and 1 aerobic exercise session per week (78 sessions total). The combination exercise sessions consisted of a brief warm-up on a treadmill or cycle ergometer (3-5 minutes), aerobic exercise, and resistance training followed by a cooldown. The aerobic exercise consisted of 30 minutes of bicycling, aerobic dancing, walking, and/or jogging. The resistance training involved completing 1 set of 8 different exercises at 8 to 12 repetitions: biceps curl, triceps curl, shoulder press, standing row, abdominal crunch, heel raise, chest press, and squat. Participants were provided an online REDCap portal to report their daily exercise activities, which consisted of attendance at the gym (7212 of 8686 exercise sessions [83%]), independent exercise (at home or while traveling) (634 of 8686 exercise sessions [7%]), or both gym and independent exercise on the same day (840 of 8686 exercise sessions [10%]). Physical activity was assessed by group session attendance, exercise resistance, exercise intensity, and exercise frequency data collected from exercise logs, surveys, and evaluations.

Outcome Measurements and Follow-up

The primary outcome of the IMPACT study was the absolute change in HbA_{1c} at 3 and 6 months across the 3 groups. HbA_{1c} was analyzed using a DCA Vantage Analyzer (Siemens Healthineers). Antidiabetes medication type and frequency were recorded at baseline and follow-up visits, and the changes in medications were classified as increased, decreased, or unchanged by 2 physicians independently. If a discrepancy was noted between the physicians, a third clinician was consulted, and classification was made by consensus.

Secondary outcomes included changes in self-reported physical activity and self-reported measures of quality of life from baseline to 6 months. Self-reported levels of overall physical activity, including physical activity associated with the intervention, were evaluated with the Modifiable Activity Questionnaire as minutes of metabolic equivalent tasks (MET-minutes) per week.[20,23] Quality of life was evaluated by the General Health domain from the Short Form Health Survey–12 Item, converted from a 5-point scale to a scale of 0 to 100, with higher scores indicating better general health.[24] Instances of self-reported physical activity exceeding 3000 MET-minutes per week (3 times US Department of Health and Human Services recommendations)[25] were excluded as outliers (80 of 692 records [11.6%]). Baseline and follow-up visits were conducted at the University Research clinic site in Palo Alto, California.

Statistical Analysis

Sample size was determined using data from a previous exercise intervention among individuals with T2D, which showed an effect size of 0.5% reduction in HbA_{1c}.[10] Assuming equal variances, a baseline mean HbA_{1c} of 6.5%, a Bonferroni adjusted $\alpha = .05/3 = .017$, and power of 80%, we calculated that 92 patients per group would be required to detect a 0.5% difference in HbA_{1c} using a 2-sample *t* test. To account for 20% attrition, we recruited 115 participants per group, resulting in a total sample size of 345.

Primary analysis followed the intention-to-treat (ITT) principle and included all participants according to their randomized intervention. We also performed sensitivity analyses in a per-protocol (PP) cohort limited to all participants who had at least 50% adherence to the structured exercise sessions for the 6-month study period and attended at least 1 follow-up visit (3 or 6 months). Continuous baseline characteristics were presented as median and 25th (Q1) and 75th (Q3) percentiles. Categorical baseline characteristics were expressed as counts and percentages. We computed absolute standardized differences to compare the baseline characteristics by study group assignment.[26] Between-group differences for baseline characteristics were assessed by protocol adherence using a χ^2 test or Fisher exact test (for small cell counts) for categorical variables and a *t* test (for normal distribution values) or Mann-Whitney *U* test (for nonnormal distribution values) for

Downloaded from jamanetwork.com by guest on 09/23/2026

continuous measures. We assessed primary and secondary outcomes adjusting for randomization strata (HbA_{1c} and age) with a mean model structure using a constrained longitudinal analysis approach (ie, assumed no difference between study groups at baseline).[27] The model included a random effect to account for correlation between repeated measures within a participant and assumed an exchangeable correlation structure. We used $P < .05$ (2-tailed) to determine whether there was any significant difference among the 3 study groups at month 6, and if significance was found, we performed pairwise comparisons using a Bonferroni adjusted $\alpha = .05/3 = .017$ to determine significance between groups.

Data analysis was performed from January to April 2022. All analyses were performed using SAS statistical software version 9.4 (SAS Institute) and R statistical software version 4.0.4 (R Project for Statistical Computing).[28]

Results

Table 1 shows baseline characteristics of the 357 participants (143 women [40.1%]; mean [SD] age, 57.4 [11.1] years) who were randomized into 1 of 3 study groups: (1) thrice-weekly exercise intervention (119 participants), (2) once-weekly exercise intervention (119 participants), or (3) UC (119 participants). Stratified block randomization yielded balance across groups, as indicated by an absolute standardized difference of less than 0.34 for all baseline demographic variables (Table 1). The median (IQR) HbA_{1c} was 7.4% (6.9%-8.2%), the median BMI (IQR) was 31.3 (28.3-37.2), and most participants (237 participants [66.4%]) had a bachelor's degree or higher. There were no significant differences in age, sex, or weight between the participants who were and were not at least 50% adherent to exercise in the thrice-weekly exercise group (eTable 1 in Supplement 2).

Figure 1 provides a detailed diagram regarding patient recruitment, enrollment, and follow-up. From October 2016 to April 2019, a total of 357 individuals with T2D not taking insulin were randomized into the study. Of those randomized, 251 participants (70.3%) completed at least 1 follow-up visit and 192 (54.0%) were included in the PP analysis (Figure 1).

Effect of Intervention on HbA_{1c}

Figure 2 depicts HbA_{1c} levels measured over 3 time points (baseline, 3-month, and 6-month follow-up) across the study groups for the ITT and PP analyses. There was no significant difference in HbA_{1c} among study groups at 3-month or 6-month follow-up in the ITT analysis, as shown in **Table 2**. Specifically, HbA_{1c} changed by −0.26% (95% CI, −0.48% to −0.03%) in the thrice-weekly exercise group and by −0.18% (95% CI, −0.41% to 0.05%) in the once-weekly exercise group compared with UC at 3 months, and by −0.23% (95% CI, −0.48% to 0.01%) in the thrice-weekly exercise group and −0.16% (95% CI, −0.41% to 0.09%) in the once-weekly exercise group compared with UC at 6 months ($P > .017$ Bonferroni adjusted α for all HbA_{1c} pairwise comparisons).

In the PP subset (62 participants [52.1%] in the once-weekly exercise group and 56 participants [47.1%] in the thrice-weekly exercise group), there were significant differences between study groups at 3-month and 6-month follow-up. Specifically, HbA_{1c} in the thrice-weekly exercise group changed by −0.35% (95% CI, −0.60% to −0.10%; $P = .005$) at 3 months and by −0.38% (95% CI, −0.65% to −0.12%; $P = .005$) at 6 months compared with UC ($P < .017$ Bonferroni adjusted α for both comparisons). HbA_{1c} in the once-weekly exercise group changed by −0.14% (95% CI, −0.39% to 0.12%) compared with UC at 6 months, although the difference was not significant. There were no significant differences observed in thrice-weekly exercise vs the once-weekly exercise study groups. Detailed statistical output is shown in Table 2.

Adherence and Follow-up

Only 123 of 238 participants (51.7%) in the exercise groups reached at least 50% adherence to exercise sessions. Median overall adherence was comparable in the once-weekly exercise group (54.5%) and thrice-weekly exercise group (47.0%) in the ITT analysis. There were no significant

Downloaded from jamanetwork.com by guest on 09/23/2026

differences in the distance from the university research clinic site between those who were 50% adherent to structured exercise sessions (offered at >60 sites) and those who were not (mean [SD], 14.8 [9.6] miles vs 15.0 [11.3] miles). However, those who did not complete a 6-month follow-up visit lived further away from the university research clinic site (mean [SD], 13.0 [9.0] miles vs 16.7 [12.1] miles; Wilcoxon 2-sample test statistic = 33362.5; P = .002). At 6 months, 175 of 357 participants (49.0%) were lost to follow-up.

Hypoglycemic Medications

Dosage of hypoglycemic medication remained unchanged from baseline to 6-month follow-up for almost 90% of the participants who completed their 6-month study visit. In the thrice-weekly exercise group, 6 participants (5.7%) reduced their dose, 94 participants (88.7%) had unchanged doses, and 6 participants (5.7%) increased their dose. In the once-weekly exercise group, 6 participants (5.8%) reduced their dose, 93 participants (89.4%) had unchanged doses, and 5 participants (4.8%) increased their dose. In the UC group, 3 participants (3.1%) reduced their dose,

Table 1. Baseline Participant Characteristics

Characteristic	Participants, No. (%) (N = 357)		
	Thrice-weekly exercise (n = 119)	Once-weekly exercise (n = 119)	Usual care (n = 119)
Age, median (IQR), y	59.3 (49.7-65.8)	58.1 (50.5-65.2)	55.8 (48.7-64.9)
Sex			
Female	53 (44.5)	46 (38.7)	44 (37.0)
Male	66 (55.5)	73 (61.3)	75 (63.0)
Race and ethnicity[a]			
African American or Black	10 (8.5)	11 (9.3)	12 (10.2)
Asian	38 (32.2)	50 (42.4)	42 (35.6)
Hispanic or Latino	17 (14.4)	17 (14.4)	16 (13.6)
Non-Hispanic White	42 (35.6)	28 (23.7)	42 (35.6)
Other race[b]	11 (9.3)	12 (10.2)	6 (5.1)
Education[a]			
High school, general educational development, or less	2 (1.7)	10 (8.4)	7 (5.9)
Some college or associate degree	28 (23.5)	38 (31.9)	34 (28.8)
Bachelor's degree	42 (35.3)	37 (31.1)	28 (23.7)
Graduate degree	47 (39.5)	34 (28.6)	49 (41.5)
Alcohol use[a]			
Never	26 (22.2)	28 (23.9)	28 (24.1)
Rarely	48 (41.0)	40 (34.2)	41 (35.3)
Once or less per week	29 (24.8)	37 (31.6)	23 (19.8)
Twice or more per week	14 (12.0)	12 (10.3)	24 (20.7)
Tobacco use[a]			
Never	92 (77.3)	89 (76.1)	77 (67.5)
Former smoker	20 (16.8)	23 (19.7)	27 (23.7)
Current smoker	7 (5.9)	5 (4.3)	10 (8.8)
Clinical measures, median (IQR)[c]			
Hemoglobin A_{1c}, % of total hemoglobin	7.2 (6.8-8.2)	7.3 (6.9-8.2)	7.4 (6.9-8.3)
Weight, kg	86.8 (75.4-106.0)	92.1 (82.0-108.1)	96.0 (84.7-111.6)
Body mass index[d]	30.9 (28.0-35.3)	30.8 (28.6-37.2)	32.8 (28.4-38.3)
Waist circumference, cm	109.0 (100.5-120.0)	110.0 (103.0-122.0)	113.5 (105.0-126.0)
Blood pressure (sitting), mm Hg			
Systolic	129 (119-138)	132 (120-143)	131 (120-141)
Diastolic	84 (78-90)	86 (79-93)	86 (79-92)
Pulse (sitting), beats per minute	76 (70-85)	77 (69-83)	75 (69-85)

SI conversion factor: To convert hemoglobin A_{1c} to proportion of total hemoglobin, multiply by 0.01.

[a] Race and ethnicity was missing or unknown for 3 participants, education level was missing for 1 participant, alcohol use was missing or declined to answer for 7 participants, and tobacco use was missing or declined to answer for 7 participants.

[b] Other refers to American Indian/Alaska Native, Pacific Islander, or multiracial.

[c] Weight was missing for 2 participants, body mass index was missing for 2 participants, waist circumference was missing for 2 participants, systolic blood pressure was missing for 2 participants, diastolic blood pressure was missing for 2 participants, and pulse was missing for 2 participants.

[d] Body mass index is calculated as weight in kilograms divided by height in meters squared.

Downloaded from jamanetwork.com by guest on 09/23/2026

87 participants (88.8%) had an unchanged dose, and 8 participants (8.2%) increased their dose. There was no significant difference in medication changes across the 3 groups.

Effect of Intervention on Self-reported Physical Activity

Table 2 shows significant differences in MET-minutes per week between study groups at both the 3-month and the 6-month follow-up for both the ITT and PP analysis. At 6 months, participants in the thrice-weekly exercise group had 461 (ITT) and 634 (PP) more MET-minutes per week compared with those in the UC group.

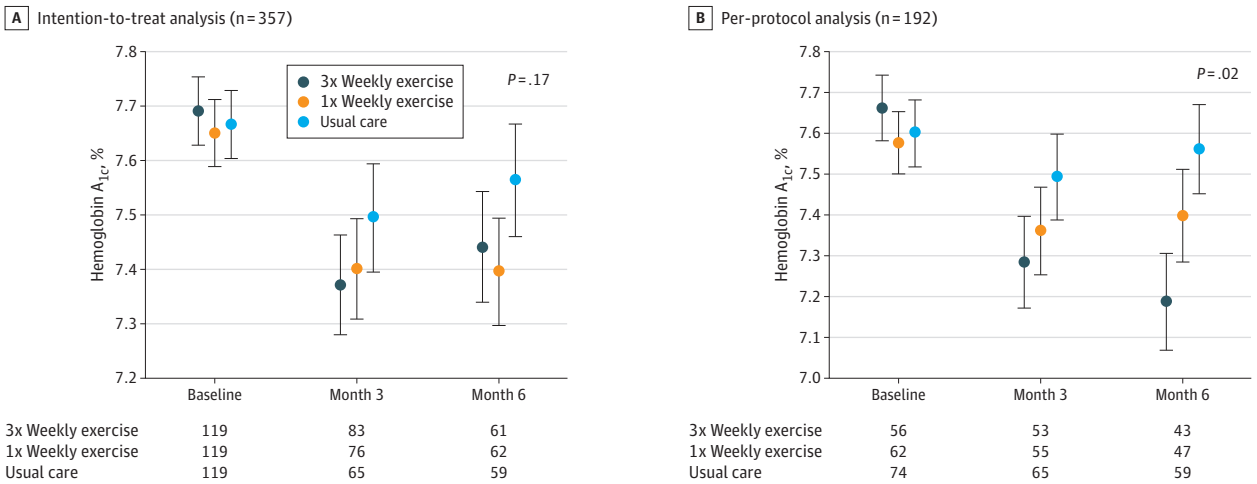
eTable 2 and eTable 3 in Supplement 2 show baseline, follow-up, and changes in clinical values in the ITT and PP populations, respectively. eTable 4 in Supplement 2 summarizes the exercise training data by month for individuals who met the per-protocol criteria. Table 2 shows no significant differences in general health in the ITT or PP analysis in any of the study groups.

Discussion

The main finding of this randomized clinical trial is that a structured exercise program offering exercise thrice or once weekly did not significantly reduce HbA_{1c} in adults with T2D compared with UC. Furthermore, overall adherence to the structured exercise program was low in both exercise groups. There was evidence of HbA_{1c} reduction in the exercise groups, although the effects in the ITT and once-weekly exercise PP analyses did not achieve prespecified levels of significance compared with UC. The thrice-weekly exercise intervention was effective in improving HbA_{1c} in individuals reaching at least 50% adherence to the structured exercise sessions (PP analysis). The once-weekly exercise intervention was not associated with a significant difference from UC in HbA_{1c} levels in the PP analysis, suggesting the importance of offering a higher number of weekly structured exercise sessions to meet ADA exercise recommendations and reduce HbA_{1c}.

Participants in the thrice-weekly exercise group with at least 50% adherence had a mean change in HbA_{1c} level (−0.38%; 95% CI, −0.65% to −0.12%) similar to that in a previous study (−0.27%; 95% CI, −0.46% to −0.08%).[10] These results suggest that a thrice-weekly structured exercise with close monitoring and support of adherence is necessary to make meaningful clinical changes in diabetes severity. These results support offering structured exercise sessions to meet the

Figure 2. Hemoglobin A_{1c} (HbA_{1c}) Estimates by Study Group



Graphs show the estimated mean HbA_{1c} from the fitted model (Table 2) for each study group at each time point (grouped by time point) along with SEs (denoted by error bars) around each point estimate.

Downloaded from jamanetwork.com by guest on 09/23/2026

Table 2. Baseline, 3-Month Follow-up, and 6-Month Follow-up Changes From Mixed-Effects Model

Outcome and study group	Intention-to-treat		Per-protocol	
	Adjusted mean (95% CI)	*P* value	Adjusted mean (95% CI)	*P* value
Hemoglobin A_{1c}, % of total hemoglobin				
Baseline				
Thrice-weekly exercise	7.69 (7.57 to 7.81)	NA	7.66 (7.50 to 7.82)	NA
Once-weekly exercise	7.65 (7.53 to 7.77)		7.58 (7.43 to 7.73)	
UC	7.67 (7.54 to 7.79)		7.60 (7.44 to 7.76)	
3 mo				
Thrice-weekly exercise	7.37 (7.19 to 7.55)	.08[a]	7.28 (7.06 to 7.51)	.02[a]
Once-weekly exercise	7.40 (7.22 to 7.58)		7.36 (7.15 to 7.57)	
UC	7.49 (7.30 to 7.69)		7.49 (7.28 to 7.70)	
6 mo				
Thrice-weekly exercise	7.44 (7.24 to 7.64)	.17[a]	7.19 (6.95 to 7.42)	.02[a]
Once-weekly exercise	7.40 (7.20 to 7.59)		7.40 (7.17 to 7.62)	
UC	7.56 (7.36 to 7.77)		7.56 (7.35 to 7.78)	
Pairwise comparisons, estimate (95% CI)				
3 mo				
Thrice-weekly exercise vs UC	−0.26 (−0.48 to −0.03)	NA[b]	−0.35 (−0.60 to −0.10)	.005[b]
Once-weekly exercise vs UC	−0.18 (−0.41 to 0.05)		−0.21 (−0.46 to 0.03)	.09[b]
Thrice-weekly exercise vs once-weekly exercise	−0.07 (−0.29 to 0.14)		−0.14 (−0.39 to 0.11)	.28[b]
6 mo				
Thrice-weekly exercise vs UC	−0.23 (−0.48 to 0.01)	NA[b]	−0.38 (−0.65 to −0.12)	.005[b]
Once-weekly exercise vs UC	−0.16 (−0.41 to 0.09)		−0.14 (−0.39 to 0.12)	.30[b]
Thrice-weekly exercise vs once-weekly exercise	−0.07 (−0.32 to 0.17)		−0.24 (−0.52 to 0.03)	.09[b]
Modifiable Activity Questionnaire score				
Baseline				
Thrice-weekly exercise	738 (560 to 916)	NA	735 (513 to 957)	NA
Once-weekly exercise	733 (557 to 908)		730 (530 to 930)	
UC	726 (494 to 958)		704 (471 to 937)	
3 mo				
Thrice-weekly exercise	1126 (855 to 1397)	.005[a]	1277 (1008 to 1546)	.003[a]
Once-weekly exercise	782 (563 to 1002)		866 (624 to 1109)	
UC	831 (536 to 1125)		811 (488 to 1135)	
6 mo				
Thrice-weekly exercise	1108 (799 to 1417)	.01[a]	1269 (926 to 1612)	<.001[a]
Once-weekly exercise	949 (692 to 1206)		983 (721 to 1246)	
UC	656 (209 to 1103)		648 (202 to 1094)	
Pairwise comparisons, estimate (95% CI)				
3 mo				
Thrice-weekly exercise vs UC	304 (63 to 545)	.01[b]	448 (173 to 723)	.001[b]
Once-weekly exercise vs UC	−63 (−310 to 183)	.61[b]	38 (−235 to 310)	.79[b]
Thrice-weekly exercise vs once-weekly exercise	367 (129 to 605)	.003[b]	410 (121 to 700)	.006[b]
6 mo				
Thrice-weekly exercise vs UC	461 (160 to 761)	.003[b]	634 (312 to 957)	<.001[b]
Once-weekly exercise vs UC	274 (−28 to 576)	.08[b]	315 (4 to 626)	.04[b]
Thrice-weekly exercise vs once-weekly exercise	187 (−112 to 486)	.22[b]	319 (−20 to 658)	.07[b]

(continued)

Downloaded from jamanetwork.com by guest on 09/23/2026

Table 2. Baseline, 3-Month Follow-up, and 6-Month Follow-up Changes From Mixed-Effects Model (continued)

Outcome and study group	Intention-to-treat		Per-protocol	
	Adjusted mean (95% CI)	*P* value	Adjusted mean (95% CI)	*P* value
Short Form Health Survey-12 Item, General Health domain score				
Baseline				
Thrice-weekly exercise	47 (44 to 50)	NA	50 (45 to 55)	NA
Once-weekly exercise	47 (44 to 50)		50 (46 to 55)	
UC	47 (44 to 50)		50 (45 to 54)	
3 mo				
Thrice-weekly exercise	57 (52 to 61)	.15[a]	61 (55 to 67)	.07[a]
Once-weekly exercise	53 (48 to 57)		56 (50 to 62)	
UC	52 (47 to 57)		53 (47 to 59)	
6 mo				
Thrice-weekly exercise	57 (52 to 62)	.25[a]	62 (56 to 69)	.05[a]
Once-weekly exercise	60 (55 to 65)		64 (58 to 70)	
UC	54 (49 to 59)		56 (50 to 62)	
Pairwise comparisons, estimate (95% CI)				
3 mo				
Thrice-weekly exercise vs UC	4.8 (−0.7 to 10.2)	NA[b]	7.1 (0.9 to 13.4)	NA[b]
Once-weekly exercise vs UC	0.4 (−5.1 to 6.0)		1.8 (−4.3 to 7.9)	
Thrice-weekly exercise vs once-weekly exercise	4.3 (−0.9 to 9.6)		5.3 (−1.1 to 11.8)	
6 mo				
Thrice-weekly exercise vs UC	2.5 (−3.5 to 8.4)	NA[b]	5.7 (−0.9 to 12.4)	NA[b]
Once-weekly exercise vs UC	5.1 (−0.9 to 11.1)		7.7 (1.2 to 14.2)	
Thrice-weekly exercise vs once-weekly exercise	−2.6 (−8.4 to 3.2)		−1.9 (−8.8 to 4.9)	

Abbreviations: NA, not applicable; UC, usual care.

SI conversion factor: To convert hemoglobin A_{1c} to proportion of total hemoglobin, multiply by 0.01.

[a] The overall *P* values correspond to a test of whether there is any difference between the interventions at each time point. The test is performed at a significance level of .05.

[b] The *P* value corresponds to a pairwise test evaluating whether there are differences between the 2 groups. The test is performed at a significance level of .017 (.05/3 per Bonferroni). The tests are only shown if the overall *P* value for the corresponding time point were <.05.

ADA exercise guidelines for individuals with T2D,[2] rather than advice alone, as also evidenced in the Activity Counseling Trial.[15]

Compared with UC, participants randomized to the thrice-weekly exercise group (the frequency recommended by ADA) had a substantial increase in their self-reported physical activity at 6 months, whereas no significant difference was observed in the once-weekly exercise group. Furthermore, there was no significant difference observed in the General Health score among the 3 study groups. Many previous studies investigating exercise in individuals with T2D have found exercise has positive effects on general well-being[22]; however, these studies used noncontrolled designs and had a small number of participants enrolled.[29] Findings in IMPACT were consistent with those of the Diabetes Aerobic and Resistance Exercise study,[29] whereas the Health Benefits of Aerobic and Resistance Training in Individuals With Type 2 Diabetes study[30] found that participants in exercise groups did improve their self-reported general health. Additionally, we did not observe improvement in participants' weight. This is consistent with other trials showing minimal weight loss in exercise-alone interventions, with the greatest weight loss observed in combined diet and exercise interventions.[31,32] Thus, structured exercise is most beneficial in combination with nutritional recommendations for individuals with T2D for lowering weight and HbA_{1c}.

Individuals with T2D are among the least likely to engage in regular physical activity, with participation rates significantly below national norms (58% of US adults are physically active compared with 39% of adults with diabetes).[33] In addition, their adherence to physical activity programs is as low as 50%.[16,34] In IMPACT, attrition was similar across study groups. There was a substantial degree of loss to follow-up at 6 months (49%), which is typically observed in general population behavioral interventions (12%-50%).[19,35] This may be because follow-up was conducted in only 1 clinic site, and we observed that participants who lived further away from this site were less

Downloaded from jamanetwork.com by guest on 09/23/2026

likely to attend a follow-up visit. Regarding adherence, since the IMPACT study did not have a run-in period as an exclusion criterion, we were able to investigate the issue of adherence in a community-based program setting: only 51.7% of participants in the exercise groups reached at least 50% adherence to exercise sessions, similar to other empirical studies. Despite the lower time commitment and greater flexibility to complete exercise on their own in the once-weekly exercise, the median overall adherence was comparable in the once-weekly exercise group (54.5%) and thrice-weekly exercise group (47.0%) in the ITT analysis. This highlights the importance of providing sufficient weekly structured exercise sessions to fulfill the ADA guidelines, rather than fewer structured exercise sessions and recommendations for independent exercise. It is well-established that quitting motivation and confidence are associated with greater adherence to nicotine replacement therapy[36]; however, self-efficacy questions assessing IMPACT participants' confidence in the intervention and belief that the intervention would be effective were ultimately not associated with exercise adherence or a reduction in HbA_{1c}. In the thrice-weekly exercise group, there were no significant differences in demographic characteristics between those who were 50% exercise adherent vs nonadherent. Adherence to exercise sessions at a recommended frequency and duration in a general population setting is a key factor in improving glycemic control.

In the future, we may consider placing follow-up centers physically closer to participants or using mobile data collection. Previous research has identified strategies to improve adherence, including incentives,[37] on-site study staff monitoring, and on-site fitness coaching,[38-41] and using psychological interventions.[42] Structural, psychological, and social support are critical to ensure adequate adherence to exercise interventions.[43,44]

Strengths and Limitations

We studied a large sample that had diverse demographic representation using a controlled, randomized study design in a general population setting, making our findings generalizable. We reported change in self-reported physical activity levels in addition to glycemic control.

A limitation of our study is that only 70.3% of participants returned for at least 1 follow-up visit; however, this is consistent with other studies assessing the effect of exercise on individuals with T2D.[16] This may be attributed to the long distance from many participants' residences to our clinic site because participants were sampled from a large metropolitan area (approximately 18 000 km^2), across 135 zip codes, and randomized from 117 zip codes.

Conclusions

In this randomized clinical trial, there was no significant difference in HbA_{1c} in the ITT analysis. Those at least 50% adherent to the thrice-weekly structured exercise program had a significant reduction in HbA_{1c}. Providing thrice-weekly structured exercise is more beneficial than providing once-weekly structured exercise or physical activity advice alone (UC). Future efforts in increasing exercise in T2D should focus on offering structured exercise at least thrice weekly, with additional support to achieve at least 50% adherence to structured exercise programs.

ARTICLE INFORMATION

Accepted for Publication: October 31, 2022.

Published: December 21, 2022. doi:10.1001/jamanetworkopen.2022.47858

Open Access: This is an open access article distributed under the terms of the CC-BY License. © 2022 Mukherji AB et al. *JAMA Network Open*.

Corresponding Author: Latha Palaniappan, MD, MS, Division of General Medical Disciplines, Department of Medicine, Stanford University School of Medicine, 1265 Welch Rd, MC: 5475, Stanford, CA 94305 (lathap@stanford.edu).

Downloaded from jamanetwork.com by guest on 09/23/2026

Author Affiliations: Division of Primary Care and Population Health, Stanford University School of Medicine, Stanford, California (Mukherji, Talamoa, Palaniappan); Quantitative Sciences Unit, Stanford University School of Medicine, Stanford, California (Lu, Qin, Hedlin, Chung); Pennington Biomedical Research Center, Louisiana State University, Baton Rouge (Johannsen); Division of Cardiovascular Medicine, Stanford University School of Medicine, Stanford, California (Kobayashi, Haddad, Lamendola); Stanford Cardiovascular Institute, Stanford, California (Kobayashi, Haddad); Division of Endocrinology, Stanford University School of Medicine, Stanford, California (Basina); Cardiology Division, VA Palo Alto Health Care System, Stanford University, Palo Alto, California (Myers); Division of General Medical Disciplines, Department of Medicine, Stanford University School of Medicine, Stanford, California (Palaniappan).

Author Contributions: Ms Mukherji and Dr Palaniappan had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Lu, Hedlin, Chung, Haddad, Palaniappan.

Acquisition, analysis, or interpretation of data: All authors.

Drafting of the manuscript: Mukherji, Lu, Qin, Hedlin, Johannsen, Chung, Talamoa.

Critical revision of the manuscript for important intellectual content: Mukherji, Lu, Qin, Hedlin, Johannsen, Chung, Kobayashi, Haddad, Lamendola, Basina, Myers, Palaniappan.

Statistical analysis: Mukherji, Lu, Qin, Hedlin, Chung, Lamendola.

Administrative, technical, or material support: Mukherji, Lu, Johannsen, Kobayashi, Haddad, Lamendola, Myers.

Supervision: Mukherji, Haddad, Palaniappan.

Conflict of Interest Disclosures: None reported.

Funding/Support: Funding for this study was supported by a grant from the National Institutes of Health (5R18DK096394). This work was partially supported by the Quantitative Sciences Unit through the Stanford Diabetes Research Center (P30DK116074).

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 3.

Additional Contributions: The authors thank all study participants who participated in this study and the study team for their tireless work. The Stanford REDCap platform (http://redcap.stanford.edu) is developed and operated by Stanford Medicine Research IT team. The REDCap platform services at Stanford are subsidized by Stanford School of Medicine Research Office and the National Center for Research Resources and the National Center for Advancing Translational Sciences, National Institutes of Health, through grant UL1 TR001085.

REFERENCES

1. Centers for Disease Control and Prevention. National Diabetes statistics report: estimates of diabetes and its burden in the United States. January 20, 2022. Accessed July 5, 2022. https://www.cdc.gov/diabetes/data/statistics-report/index.html

2. American Diabetes Association. 5. Lifestyle Management: *Standards of Medical Care in Diabetes-2019. Diabetes Care*. 2019;42(suppl 1):S46-S60. doi:10.2337/dc19-S005

3. Centers for Disease Control and Prevention. Preventing diabetes-related complications. December 30, 2021. Accessed July 5, 2022. https://www.cdc.gov/diabetes/data/statistics-report/preventing-complications.html

4. Castaneda C, Layne JE, Munoz-Orians L, et al. A randomized controlled trial of resistance exercise training to improve glycemic control in older adults with type 2 diabetes. *Diabetes Care*. 2002;25(12):2335-2341. doi:10.2337/diacare.25.12.2335

5. Boulé NG, Haddad E, Kenny GP, Wells GA, Sigal RJ. Effects of exercise on glycemic control and body mass in type 2 diabetes mellitus: a meta-analysis of controlled clinical trials. *JAMA*. 2001;286(10):1218-1227. doi:10.1001/jama.286.10.1218

6. Baldi JC, Snowling N. Resistance training improves glycaemic control in obese type 2 diabetic men. *Int J Sports Med*. 2003;24(6):419-423. doi:10.1055/s-2003-41173

7. Gordon BA, Benson AC, Bird SR, Fraser SF. Resistance training improves metabolic health in type 2 diabetes: a systematic review. *Diabetes Res Clin Pract*. 2009;83(2):157-175. doi:10.1016/j.diabres.2008.11.024

8. Tresierras MA, Balady GJ. Resistance training in the treatment of diabetes and obesity: mechanisms and outcomes. *J Cardiopulm Rehabil Prev*. 2009;29(2):67-75. doi:10.1097/HCR.0b013e318199ff69

Downloaded from jamanetwork.com by guest on 09/23/2026

9. Richter EA, Jensen P, Kiens B, Kristiansen S. Sarcolemmal glucose transport and GLUT-4 translocation during exercise are diminished by endurance training. *Am J Physiol*. 1998;274(1):E89-E95. doi:10.1152/ajpendo.1998.274.1.E89

10. Church TS, Blair SN, Cocreham S, et al. Effects of aerobic and resistance training on hemoglobin A_{1c} levels in patients with type 2 diabetes: a randomized controlled trial. *JAMA*. 2010;304(20):2253-2262. doi:10.1001/jama.2010.1710

11. Sigal RJ, Kenny GP, Boulé NG, et al. Effects of aerobic training, resistance training, or both on glycemic control in type 2 diabetes: a randomized trial. *Ann Intern Med*. 2007;147(6):357-369. doi:10.7326/0003-4819-147-6-200709180-00005

12. Lambers S, Van Laethem C, Van Acker K, Calders P. Influence of combined exercise training on indices of obesity, diabetes and cardiovascular risk in type 2 diabetes patients. *Clin Rehabil*. 2008;22(6):483-492. doi:10.1177/0269215508084582

13. Balducci S, Leonetti F, Di Mario U, Fallucca F. Is a long-term aerobic plus resistance training program feasible for and effective on metabolic profiles in type 2 diabetic patients? *Diabetes Care*. 2004;27(3):841-842. doi:10.2337/diacare.27.3.841

14. Umpierre D, Ribeiro PA, Kramer CK, et al. Physical activity advice only or structured exercise training and association with HbA_{1c} levels in type 2 diabetes: a systematic review and meta-analysis. *JAMA*. 2011;305(17):1790-1799. doi:10.1001/jama.2011.576

15. Writing Group for the Activity Counseling Trial Research Group. Effects of physical activity counseling in primary care: the Activity Counseling Trial—a randomized controlled trial. *JAMA*. 2001;286(6):677-687. doi:10.1001/jama.286.6.677

16. Bullard T, Ji M, An R, Trinh L, Mackenzie M, Mullen SP. A systematic review and meta-analysis of adherence to physical activity interventions among three chronic conditions: cancer, cardiovascular disease, and diabetes. *BMC Public Health*. 2019;19(1):636. doi:10.1186/s12889-019-6877-z

17. MacDonald CS, Ried-Larsen M, Soleimani J, et al. A systematic review of adherence to physical activity interventions in individuals with type 2 diabetes. *Diabetes Metab Res Rev*. 2021;37(8):e3444. doi:10.1002/dmrr.3444

18. Nam S, Dobrosielski DA, Stewart KJ. Predictors of exercise intervention dropout in sedentary individuals with type 2 diabetes. *J Cardiopulm Rehabil Prev*. 2012;32(6):370-378. doi:10.1097/HCR.0b013e31826be485

19. Yamaguchi S, Waki K, Tomizawa N, et al. Previous dropout from diabetic care as a predictor of patients' willingness to use mobile applications for self-management: a cross-sectional study. *J Diabetes Investig*. 2017;8(4):542-549. doi:10.1111/jdi.12613

20. Faroqi L, Wong M, Bonde S, et al. Evaluating the clinical implementation of structured exercise: a randomized controlled trial among non-insulin dependent type II diabetics. *Contemp Clin Trials*. 2018;74:25-31. doi:10.1016/j.cct.2018.09.003

21. Schulz KF, Altman DG, Moher D; CONSORT Group. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. *Int J Surg*. 2011;9(8):672-677. doi:10.1016/j.ijsu.2011.09.004

22. Colberg SR, Sigal RJ, Yardley JE, et al. Physical activity/exercise and diabetes: a position statement of the American Diabetes Association. *Diabetes Care*. 2016;39(11):2065-2079. doi:10.2337/dc16-1728

23. Kriska A. Modifiable activity questionnaire. *Med Sci Sports Exerc*. 1997;29(suppl)(6):S73-S78.

24. Ware J Jr, Kosinski M, Keller SDA. A 12-Item Short-Form Health Survey: construction of scales and preliminary tests of reliability and validity. *Med Care*. 1996;34(3):220-233. doi:10.1097/00005650-199603000-00003

25. US Department of Health and Human Services. 2008 Physical activity guidelines for Americans. October 2008. Accessed November 14, 2022. https://www.health.gov/paguidelines/guidelines/default.aspx

26. Austin PC. Using the standardized difference to compare the prevalence of a binary variable between two groups in observational research. *Commun Stat Simul Comput*. 2009;38(6):1228-1234. doi:10.1080/03610910902859574

27. Liang KY, Zeger SL. Longitudinal data analysis of continuous and discrete responses for pre-post designs. *Indian J Stat*. 2000;62(1):134-148. Accessed November 14, 2022. https://www.jstor.org/stable/25053123

28. R Core Team. R: a language and environment for statistical computing. R Foundation for Statistical Computing. 2021. Accessed November 14, 2022. https://www.R-project.org/

29. Reid RD, Tulloch HE, Sigal RJ, et al. Effects of aerobic exercise, resistance exercise or both, on patient-reported health status and well-being in type 2 diabetes mellitus: a randomised trial. *Diabetologia*. 2010;53(4):632-640. doi:10.1007/s00125-009-1631-1

Downloaded from jamanetwork.com by guest on 09/23/2026

30. Myers VH, McVay MA, Brashear MM, et al. Exercise training and quality of life in individuals with type 2 diabetes: a randomized controlled trial. *Diabetes Care*. 2013;36(7):1884-1890. doi:10.2337/dc12-1153

31. Foster-Schubert KE, Alfano CM, Duggan CR, et al. Effect of diet and exercise, alone or combined, on weight and body composition in overweight-to-obese postmenopausal women. *Obesity (Silver Spring)*. 2012;20(8):1628-1638. doi:10.1038/oby.2011.76

32. Slentz CA, Bateman LA, Willis LH, et al. Effects of exercise training alone vs a combined exercise and nutritional lifestyle intervention on glucose homeostasis in prediabetic individuals: a randomised controlled trial. *Diabetologia*. 2016;59(10):2088-2098. doi:10.1007/s00125-016-4051-z

33. Morrato EH, Hill JO, Wyatt HR, Ghushchyan V, Sullivan PW. Physical activity in U.S. adults with diabetes and at risk for developing diabetes, 2003. *Diabetes Care*. 2007;30(2):203-209. doi:10.2337/dc06-1128

34. Kemps H, Kränkel N, Dörr M, et al. Exercise training for patients with type 2 diabetes and cardiovascular disease: what to pursue and how to do it—a position paper of the European Association of Preventive Cardiology (EAPC). *Eur J Prev Cardiol*. 2019;26(7):709-727. doi:10.1177/2047487318820420

35. Dunkley AJ, Bodicoat DH, Greaves CJ, et al. Diabetes prevention in the real world: effectiveness of pragmatic lifestyle interventions for the prevention of type 2 diabetes and of the impact of adherence to guideline recommendations: a systematic review and meta-analysis. *Diabetes Care*. 2014;37(4):922-933. doi:10.2337/dc13-2195

36. Kim N, McCarthy DE, Loh WY, et al. Predictors of adherence to nicotine replacement therapy: machine learning evidence that perceived need predicts medication use. *Drug Alcohol Depend*. 2019;205:107668. doi:10.1016/j.drugalcdep.2019.107668

37. Mitchell MS, Goodman JM, Alter DA, et al. Financial incentives for exercise adherence in adults: systematic review and meta-analysis. *Am J Prev Med*. 2013;45(5):658-667. doi:10.1016/j.amepre.2013.06.017

38. Broekhuizen K, de Gelder J, Wijsman CA, et al. An internet-based physical activity intervention to improve quality of life of inactive older adults: a randomized controlled trial. *J Med Internet Res*. 2016;18(4):e74. doi:10.2196/jmir.4335

39. Sullivan AN, Lachman ME. Behavior change with fitness technology in sedentary adults: a review of the evidence for increasing physical activity. *Front Public Health*. 2017;4:289. doi:10.3389/fpubh.2016.00289

40. Martin SS, Feldman DI, Blumenthal RS, et al. mActive: a randomized clinical trial of an Automated mHealth Intervention for Physical Activity Promotion. *J Am Heart Assoc*. 2015;4(11):e002239. doi:10.1161/JAHA.115.002239

41. Spring B, Schneider K, McFadden HG, et al. Multiple behavior changes in diet and activity: a randomized controlled trial using mobile technology. *Arch Intern Med*. 2012;172(10):789-796. doi:10.1001/archinternmed.2012.1044

42. Martinus R, Corban R, Wackerhage H, Atkins S, Singh J. Effect of psychological intervention on exercise adherence in type 2 diabetic subjects. *Ann N Y Acad Sci*. 2006;1084(1):350-360. doi:10.1196/annals.1372.024

43. Qiu SH, Sun ZL, Cai X, Liu L, Wang B. Improving patients' adherence to physical activity in diabetes mellitus: a review. *Diabetes Metab J*. 2012;36(1):1-5. doi:10.4093/dmj.2012.36.1.1

44. Collado-Mateo D, Lavín-Pérez AM, Peñacoba C, et al. Key factors associated with adherence to physical exercise in patients with chronic diseases and older adults: an umbrella review. *Int J Environ Res Public Health*. 2021;18(4):2023. doi:10.3390/ijerph18042023

SUPPLEMENT 1.

Trial Protocol and Statistical Analysis Plan

SUPPLEMENT 2.

eTable 1. Comparison of Baseline Participant Characteristics: 50% Exercise Adherence vs Nonadherence by Study Group (Per-Protocol Population)
eTable 2. Baseline, Follow-up, and Changes in Clinical Values in the ITT Population
eTable 3. Baseline, Follow-up, and Changes in Clinical Values in the PP Population
eTable 4. Exercise Intervention Data for Individuals Who Met the Per-Protocol Criteria (N = 192)

SUPPLEMENT 3.

Data Sharing Statement

Downloaded from jamanetwork.com by guest on 09/23/2026